

Ocean Physics & Structure

Mixed layer, barrier layer, thermocline and heat content - computed with real seawater physics, and refused outright where the inputs cannot support them.

PORT	PAGE FILE	MODULES	EQUATION OF STATE
8505	app/phase2/physics_page.py	phase2/physics/{seawater, layers, ohc}.py	EOS-80

A temperature profile is raw material. What an oceanographer, a fisherman or a cyclone forecaster actually needs are the structures inside it: how deep the well-mixed surface layer goes, where the sharp temperature drop sits, and how much heat the column is storing. This page computes those, and says no when it cannot.

1. What this feature is, in plain words

The **mixed layer** is the top of the ocean that wind and waves have stirred into a single uniform slab. Its depth controls how quickly the surface warms, how nutrients reach the light, and how much heat a storm can draw on.

The **thermocline** is the sharp drop underneath it, where temperature falls fastest. The **barrier layer** is a North Indian Ocean speciality: fresh river water sits on top of saltier water and forms a lid that stops mixing, which is one reason the Bay of Bengal intensifies cyclones so effectively. **Ocean heat content** is the total heat stored down to a chosen depth.

All of these are computed with genuine seawater physics rather than rules of thumb. The equation of state used to turn temperature and salinity into density was checked against four published reference values and agrees to better than one thousandth of a kilogram per cubic metre - including the standard check value, 1023.343.

And here is the part that a jury should be told deliberately: **the page refuses to compute two of these from satellite input**. Density needs salinity at depth, and our satellite-driven model predicts temperature only. The reanalysis salinity is sitting right there in the same file and would produce a plausible-looking number - so the page shows a 'refused, not approximated' box instead, and a test asserts the code neither reads nor returns that salinity.

THE EVERYDAY PICTURE

Temperature alone is like knowing a patient's temperature at fifteen points down their body. Structure is the diagnosis: where the layers are, how thick the insulating one is, and how much energy is stored. This page is the diagnosis step.

2. What you actually see on the screen

One date, one grid, three selectable sources:

- **Mixed layer depth** by the density criterion - a 0.03 kg/m³ change from 10 m, the standard published definition.
- **Isothermal layer depth**, its temperature-only counterpart, and the difference between the two, which is the barrier layer.
- **Thermocline depth** - where the vertical temperature gradient is steepest.
- **Ocean heat content**, computed with real density from temperature and salinity where the source allows it.
- A source toggle - GLORYS reanalysis, or the shipped satellite model - reading the same day from the same bundle, so it is a true model-versus-truth comparison and not two different eras.
- A live bias table with a warning printed above the maps when a comparison is not defensible.

- A vertical-gradient consistency panel showing how faithfully the model reproduces the real shape of the temperature drop.

3. How to run it, step by step

RUN THIS

```
PYTHONPATH=src python -m streamlit run app/phase2/physics_page.py --server.port 8505
```

- 1 Run the command and open **http://localhost:8505**.
- 2 Leave the source on **glorys** first and pick a date. All four structure fields appear, computed with real density.
- 3 Switch the source to **v2 satellite**. Thermocline and isothermal layer depth still compute; mixed layer by density and barrier layer are replaced by an explicit refusal box. **Stop and read that box aloud** - it is one of the strongest honesty moments in the demo.
- 4 Open the bias table underneath. It is computed live on the date the viewer chose, not quoted from a stored file.
- 5 Finish on the gradient-consistency panel, which shows the model reproducing the thermocline shape at essentially 100% of the observed gradient.

MEASURE THE GRADIENT CONSISTENCY YOURSELF

```
PYTHONPATH=src python scripts/phase2/measure_physical_consistency.py
```

4. Why the problem statement needs this

The problem statement is sponsored by a ministry whose users are forecasters and marine operators. Those users do not act on a temperature at 75 m; they act on mixed layer depth, on heat content, on where the thermocline sits. This feature is the translation layer between a research output and an operational one.

It is also where the project's compliance boundary is enforced in code rather than in prose. The requirement is a satellite-input system. Quietly borrowing reanalysis salinity to make a prettier map would break that requirement while looking like a feature.

PS requirement	How this feature answers it
Satellite-observation inputs only	Refuses any structure field that would need reanalysis salinity, and a test asserts the refusal is real.
Depth-resolved output that is useful downstream	Turns 15 temperatures into the four structural quantities operational users actually request.
Physically sound treatment	EOS-80 density and Mackenzie sound speed, both pinned to published reference values by tests.

5. The measured numbers behind it

Two independent measurements, both run on this machine [VERIFIED]:

Measurement	Result	Reading
Vertical gradient, model vs independent Argo (ratio pred/obs)	8 m: 0.42 25 m: 0.69 40 m: 0.81 75-125 m: 1.006 850 m: 1.00	The thermocline is not smoothed - the model reproduces 100.6% of the observed gradient there. The flattening is in the top ~30 m only.
Stage-2 structure fields vs GLORYS (bias / RMSE, metres)	MLD -14.12 / 21.34 barrier +8.88 / 19.00 ILD -5.85 / 16.44 thermocline -1.80 / 24.12	Two of these are not trustworthy, and the page says so on screen.
Stage-2 heat content 0-300 m vs GLORYS	bias -0.024 GJ/m2, RMSE 0.664	Heat content survives. An integral is far less sensitive than a threshold crossing.

- The cause of the mixed-layer failure was traced rather than guessed: surface salinity carries +0.19 psu of bias, which is roughly 0.15 kg/m3 of density - about **five times** the 0.03 kg/m3 threshold the mixed-layer criterion uses. A threshold crossing five times smaller than the bias will trip at the wrong depth systematically.
- The standing instruction inside the project is explicit: do not use the stage-2 mixed layer or barrier layer for any claim; its heat content is defensible.
- A physics-based gradient penalty was added to the training objective and tested across three seeds. **It did not help** - and the gradient measurement above explains why: it was designed to protect structure that was never being lost.

6. The impact - and who it is for

Who	What changes for them
Cyclone forecasters	Mixed layer and barrier layer are direct inputs to intensification forecasting - the Bay of Bengal barrier layer is a known amplifier.
Fisheries	The thermocline sets where many commercially important species concentrate; its depth is an operational planning variable.
Climate monitoring	Ocean heat content is the primary measure of how much heat the planet is absorbing, and this produces it daily from satellites.
The jury	A visible, code-enforced compliance boundary - the page refuses to produce a number rather than approximate it.

WHY SOCIETY GAINS FROM THIS

Barrier layers in the Bay of Bengal are part of why cyclones there intensify so dangerously close to one of the most densely populated coastlines on Earth. Producing these structural fields daily, from satellites, with the untrustworthy ones clearly marked, is directly relevant to coastal warning.

7. Say this to the jury

A temperature profile is raw material; this turns it into the structures forecasters actually use - mixed layer, barrier layer, thermocline and heat content, with real seawater physics checked against published reference values. And notice what it does not do: on satellite input it refuses to compute mixed layer by density, because that needs salinity at depth that our satellite model does not predict. The reanalysis salinity is in the very same file and would have produced a beautiful map. We show a refusal box instead, and a test enforces it.

The refusal box is the strongest single moment in the whole demo. Do not rush it.

8. Questions the jury will ask

Q. Why not just use the reanalysis salinity to fill the gap?

A. Because the deliverable is a satellite-input system, and that would put a reanalysis field inside a number labelled satellite. It would look like a feature and function as a violation. It is a compliance boundary, not a missing capability.

Q. You said your stage-2 mixed layer is biased by 14 metres. Is that not a serious failure?

A. It is a measured failure and we publish it. We traced the cause: +0.19 psu of surface salinity bias against a 0.03 kg/m³ threshold. That is why the standing recommendation inside the project is not to use that field, while the heat content from the same model - bias 0.024 GJ/m² - is defensible. Integrals tolerate bias; threshold crossings do not.

Q. Did adding physics to the loss function improve the model?

A. No, and we measured it on three seeds rather than one. At one weight it was clearly worse; at another the mean sat inside the seed noise and the sign did not hold. We then measured why: the thermocline gradient was already at 100.6% of the observed value, so the term was protecting structure that was never being lost. A null result, reported as one.

9. The honest limit - say it before they find it

NOTHING PREDICTS SALINITY AT DEPTH FROM SATELLITES

The seven satellite channels carry a surface salinity only. That single limit is why two of the four structure fields are refused in satellite mode, and it is a sensor limitation rather than a modelling gap. Separately, the seasonal barrier-layer comparison shown on this page still rests on the Phase-1 2019-2022 monthly record and says so, because a four-year seasonal signal cannot be recomputed on a 388-day bundle without silently changing its magnitude.

IN ONE LINE

Turns fifteen temperatures into the four structures forecasters use - and refuses two of them on satellite input rather than borrowing reanalysis salinity.

100.6%

OF THE OBSERVED THERMOCLINE GRADIENT
REPRODUCED

2 fields

REFUSED RATHER THAN APPROXIMATED

-0.024 GJ/m²

HEAT-CONTENT BIAS: THE FIELD THAT
SURVIVES

Feature 5 of 17. Next: Feature 6.